

Worksheet

Stem-and-leaf plots 2

The heights of 28 students in a Year 7 class were measured (in cm). Here are the results:

182 166 166 172* 168* 161 175*

157 164 155 161* 173* 167 181

166* 168 165 165* 158* 163* 154

166 170* 181* 158 165 170* 170

(The 12 girls in the class are shown by *.)

These measurements can be grouped and listed in a **stem-and-leaf plot**.

Stem	Leaf
15	4 5 7 8 8
16	1 1 3 4 5 5 5 6 6 6 6 7 8 8
17	0 0 0 2 3 5
18	1 1 2

It groups the scores into 'classes'; most heights are in the class '160–169'. An ordered stem-and-leaf plot is useful for organising a large number of scores. It is more powerful than a column graph for statistical calculations.

- Use the stem-and-leaf plot to find:
 - the modal height _____
 - the median height _____
 - the range _____
 - the mean (correct to two decimal places) _____
 - How many students had a height of *over* 171 cm? _____
- Suppose we wish to compare the heights of boys and girls. We can do this using a **back-to-back** stem-and-leaf plot. Complete the *ordered* plot below.

Girls		Boys
	8	15 4 5 7 8
8 6 5 3 1	16	1 4
	17	
	18	

Note: Leaves branch outwards from the stem.

- Just by looking at the distribution of scores, can you see which group:
 - was taller, on average? _____
 - had a greater spread of heights? _____

- 4 Estimate (guess) the answers to complete the following table.

	Girls	Boys
Median height		
Mean height		

- 5 Find the following measures (mean correct to one decimal place).

	Girls	Boys
Modal height		
Median height		
Mean height		
Range of heights		

- 6 a How many boys were shorter than 165 cm? _____
b How many girls were taller than 165 cm? _____
- 7 Write a brief report on what you have discovered about the differences between boys' and girls' heights in the Year 7 class.

- 8 Measure the heights of the students in your class and display the data in a back-to-back stem-and-leaf plot. Complete the table below and write a report on your conclusions.

	Girls	Boys
Modal height		
Median height		
Mean height		
Range of heights		

Answers

- 1 **a** 166 cm **b** 166 cm **c** 28 cm **d** 166.68 cm

2	Girls						Boys							
				8	15	4	5	7	8					
8	6	5	3	1	16	1	4	5	5	6	6	7	8	
5	3	2	0	0	17	0								
				1	18	1	2							

- 3 **a** Girls **b** Boys

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	Girls	Boys
Mode	170	166
Median	169	165.5
Mean	168.5	165.3
Range	23	28

- 6 **a** 6 **b** 3
- 7 Girls taller on average; boys greater spread
- 8 Teacher to check